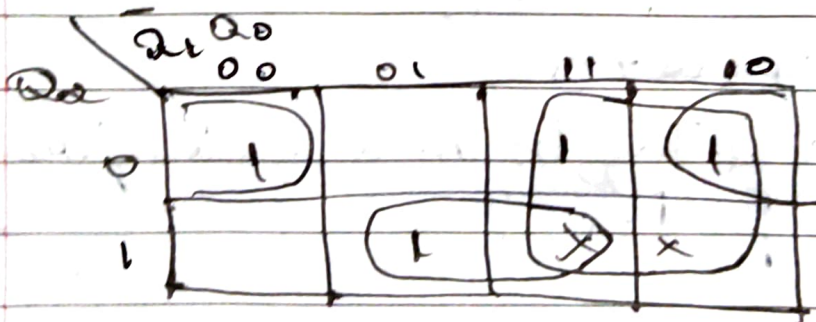


## Assignment - Day 5

Q) 1) Design a counter with T-FF whose Count sequence is 0, 2, 4, 5, 3, 1, 0, ...

|     | Present state<br>( $Q_2, Q_1, Q_0$ ) |   |   | Next state<br>( $Q_2^+, Q_1^+, Q_0^+$ ) |   |   | $T_2$ | $T_1$ | $T_0$ |
|-----|--------------------------------------|---|---|-----------------------------------------|---|---|-------|-------|-------|
| (0) | 0                                    | 0 | 0 | 0                                       | 1 | 0 | 0     | 1     | 0     |
| (2) | 0                                    | 1 | 0 | 1                                       | 0 | 0 | 1     | 1     | 0     |
| (4) | 1                                    | 0 | 0 | 1                                       | 0 | 1 | 0     | 0     | 1     |
| (5) | 1                                    | 0 | 1 | 0                                       | 1 | 1 | 1     | 1     | 0     |
| (3) | 0                                    | 1 | 1 | 0                                       | 0 | 1 | 0     | 1     | 0     |
| (1) | 0                                    | 0 | 1 | 0                                       | 0 | 0 | 0     | 0     | 1     |



$$T_1 = \overline{Q_2} \overline{Q_0} + Q_1 + Q_2 Q_0$$

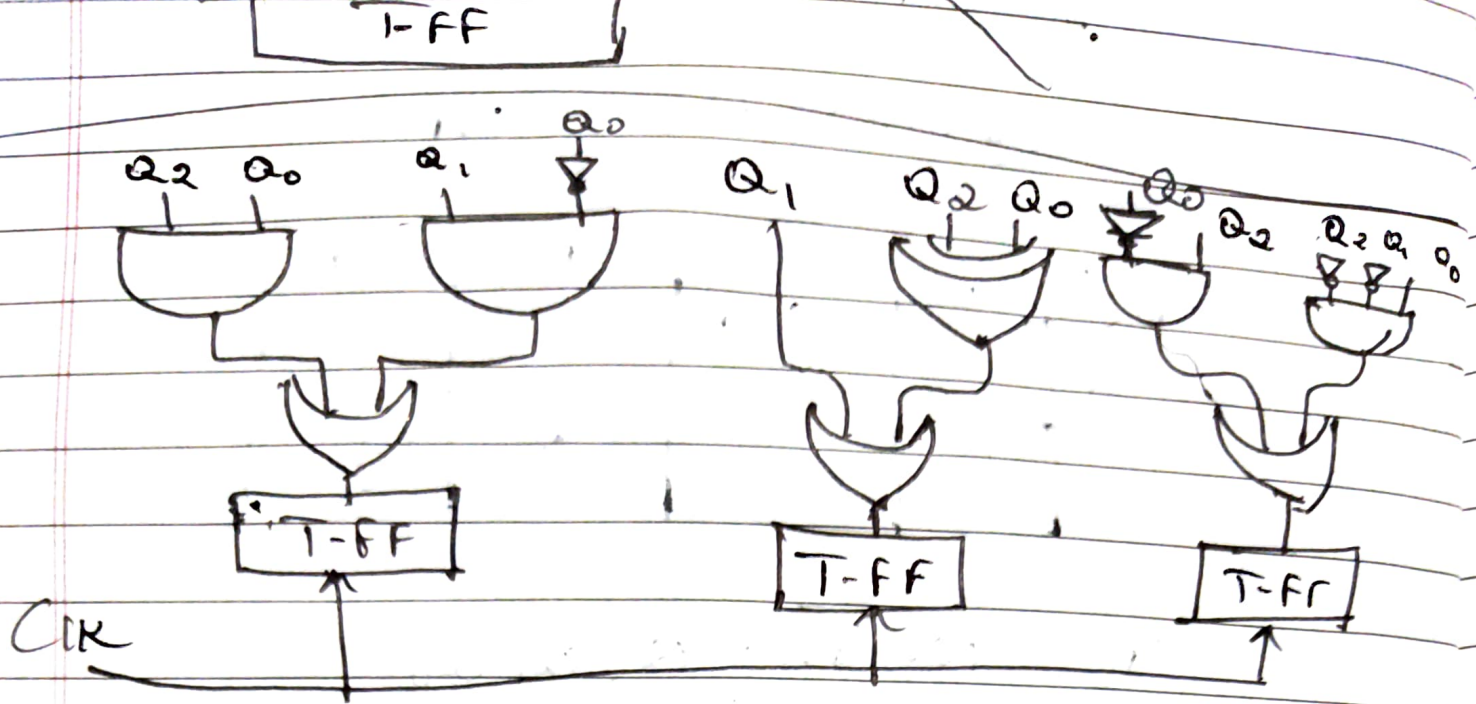
$$= Q_1 + (\overline{Q_2} \oplus Q_0)$$

| $Q_2 \backslash Q_1 Q_0$ | 00 | 01 | 11 | 10 |
|--------------------------|----|----|----|----|
| 0                        |    |    |    | 1  |
| 1                        |    | 1  | X  | X  |

$$\overline{T_2} = Q_2 Q_0 + \overline{Q_1} Q_0$$

| $Q_2 \backslash Q_1 Q_0$ | 00 | 01 | 11 | 10 |
|--------------------------|----|----|----|----|
| 0                        |    | 1  |    |    |
| 1                        | 1  |    | X  | X  |

$$\begin{aligned} T_0 &= Q_2 \overline{Q_0} + \overline{Q_2} \overline{Q_1} Q_0 \\ &= \overline{Q_0} Q_2 + Q_0 \overline{Q_1} \overline{Q_2} \end{aligned}$$



Q)2 Design a 3 bit up/down counter using Negative edge Triggered FF

★

Up Count

Down Count

Present state Next state

P.S

N.S

0 0 0

0 0 1

0 0 0

1 1 1

0 0 1

0 1 0

0 0 1

0 0 0

0 1 0

0 1 1

0 1 0

0 0 1

0 1 1

1 0 0

0 1 1

0 1 0

1 0 0

1 0 1

1 0 0

0 1 1

1 0 1

1 1 0

1 0 1

0 1 0

1 1 0

1 1 1

1 1 0

0 0 1

1 1 1

0 0 0

1 1 1

0 0 0

$$\left. \begin{array}{l} T_0 = 1 \\ T_1 = Q_0 \end{array} \right\} \text{Up Count}$$

$$T_1 = \overline{Q_0} \rightarrow \text{Down Count}$$

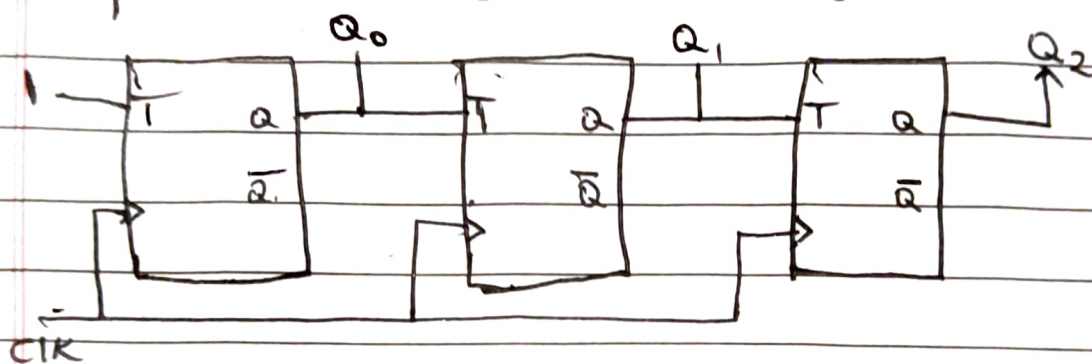
$$T_1 = \overline{UD} \cdot Q_0 + UD \cdot \overline{Q_0}$$

$$T_2 = Q_1 Q_0 \rightarrow \text{Up Count}$$

$$T_2 = \overline{Q_1} \cdot \overline{Q_0} \rightarrow \text{Down Count}$$

$$T_2 = UD(Q_1 Q_0) + \overline{UD}(\overline{Q_1} \cdot \overline{Q_0})$$

3) The circuit in the figure below is a counter what is the sequence that this circuit counts in?



Ans:  $T_0 = 1$ ,  $T_1 = Q_0$ ,  $T_2 = Q_1$

Synchronous Counter

$$Q_0^+ = Q_0 \oplus 1 = \overline{Q_0}$$

$$Q_1^+ = Q_1 \oplus Q_0$$

$$Q_2^+ = Q_2 \oplus Q_1$$

P.S

000

001

010

111

N.S

001

010

111

000

0  $\rightarrow$  1  $\rightarrow$  2  $\rightarrow$  7  $\rightarrow$  0 ...